

# The Solland Paradox

An AI-specific rebound hypothesis

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# Abstract

An efficiency improvement can reduce the physical resources required for a defined AI service while encouraging enough additional use to increase aggregate consumption. This paper calls its AI-specific formulation the “Solland Paradox.” The name denotes a proposed analytical framing, not a newly discovered economic law or an empirically established result.

The contribution is a decomposition into two demand margins: the number of tasks using AI and the quality-adjusted intensity of AI service per task. A simple resource identity states the condition under which their combined growth exceeds an efficiency gain. A measurement protocol then distinguishes that accounting condition from a causal claim that efficiency induced the extra demand.

## 1. Relation to established rebound economics

The underlying principle is the familiar rebound mechanism associated with Jevons: lower effective resource requirements can make more consumption attractive. Rebound may offset part of an engineering saving, offset all of it, or exceed it. The last case is commonly called backfire. The distinction between partial rebound and backfire is developed in the established literature. [4]

This paper proposes no replacement for that literature. Its terminology is a working label for an AI application. It claims no priority over prior analyses of technology-enabled demand expansion. McKinsey explicitly discusses an AI application of Jevons in its July 2026 analysis of data-center power architecture. [1] Prior AI research also examines Jevons and indirect environmental effects; this paper should be read within that existing discussion. [5]

The empirical hypothesis is narrower than “AI usage grows.” It is that an identifiable efficiency improvement lowers effective service costs and induces enough additional, quality-adjusted demand to exceed the corresponding physical savings.

## 2. An illustrative example

Suppose a defined task uses 100 units of a fixed physical resource. An engineering improvement reduces this to 50 units at the same output quality.

If the original demand remains one task, consumption falls to 50. If total demand becomes four equivalent tasks, consumption reaches 200. Resource use per task has halved while aggregate use has doubled.

These numbers illustrate an accounting possibility. They are not observed data. To call the increase an efficiency-induced rebound, an empirical study must establish how much of the additional demand resulted from the improvement, rather than unrelated adoption or changes in the service.

### 3. Extensive and intensive demand margins

The extensive margin is the number of tasks served. Lower effective costs may bring new users, workflows or machine-initiated tasks into scope.

The intensive margin is useful AI service per task. A workflow may demand more search, verification, personalization or reasoning. It may also use fewer calls because the improved system solves the task sooner. The direction is an empirical question.

Task intensity must be quality-adjusted output, not simply raw tokens or computation. Otherwise counting more computation as “more intelligence” makes the proposed explanation circular. Studies should predefine task families and success criteria, such as validated completions at a stated accuracy and latency.

Machine-generated activity is relevant only when recorded without double counting. A multi-step agent workflow should not simultaneously be counted as one final task and many equivalent final tasks unless the aggregation rule explicitly distinguishes them.

### 4. A resource identity and a conditional test

Choose one physical resource and a defined population over a stated period. Let  $N$  be the number of tasks,  $I$  the average quality-adjusted service units per task,  $Q = N \times I$  the total service demand,  $E$  the service units produced per unit of that resource, and  $R$  the resource consumed.

The accounting identity is:

$$R = Q / E = N \times I / E$$

For two comparable observations, resource use increases if:

$$(N_1 / N_0) \times (I_1 / I_0) > E_1 / E_0$$

Equivalently, the exact log-change identity is:

$$\Delta \ln R = \Delta \ln N + \Delta \ln I - \Delta \ln E$$

Along a specified differentiable response path, define  $\epsilon_N = d \ln N / d \ln E$  and  $\epsilon_I = d \ln I / d \ln E$ . Then:

$$d \ln R / d \ln E = \epsilon_N + \epsilon_I - 1$$

Backfire along that path requires  $\epsilon N + \epsilon I > 1$ . Equality means no aggregate resource change; a sum below one means resource consumption falls as efficiency improves. Positive demand growth below that threshold can still represent partial rebound.

These expressions are identities and conditional comparisons, not an estimated behavioral model. If E is constructed from the same Q and R observations, satisfying the identity provides no independent confirmation of causality. Measurement error, changing task mix and aggregation can distort the estimated margins.

## 5. Efficiency is not price

Physical efficiency and customer price are separate variables. Producing more useful output per joule need not generate an equal proportional price reduction. Power costs, margins, capacity constraints and product bundling affect pass-through.

Only under additional assumptions can a conventional price elasticity be interpreted as an efficiency response. If service price is inversely proportional to E, other conditions are held constant and quality-adjusted demand depends on that price, its response to efficiency follows the corresponding price response. Without those assumptions, the two elasticities differ.

Likewise, dollars, GPU-seconds and joules are separate denominators. A decline in price per completion does not by itself establish an improvement in energy efficiency. More total spending with a lower unit price does not prove physical backfire.

## 6. Why AI is a useful setting for the hypothesis

AI services can expose both margins within a single workflow. Lower effective costs may expand the task frontier while enabling more effort on each task. Verification loops, continuous monitoring and machine-initiated work are plausible channels.

There are countervailing channels. Improved reliability can reduce retries. Smaller models may replace larger ones. Better routing can avoid unnecessary work. User attention, workflow demand and budget constraints can saturate. Compute scarcity or energy prices can limit pass-through to consumers.

The hypothesis therefore requires a measured response rather than an assumption that demand is unlimited. It may hold for one resource, service or time interval and fail for another.

## 7. Forecasts motivate the question; they do not test it

The IEA's projected growth in data-center electricity demand and industry forecasts of larger AI workloads make the question relevant. [2, 3] Those projections include changing adoption, investment and service mix. They do not isolate an efficiency shock.

McKinsey's discussion of Jevons is similarly a proposed explanatory mechanism, not a causal estimate of the condition derived above. [1] Forecasts consistent with growing consumption cannot establish how consumption would have evolved without efficiency improvement.

Inference is a useful study setting because usage can recur within identifiable services. Training and retraining also recur and may have different demand responses. Results should not be transferred between them without evidence.

## 8. A credible empirical protocol

First specify the boundary: service, users, task families, resource, quality standard and time window. Publish the aggregation rules. Record task mix and changes in model or hardware configuration.

Then collect independent measures of engineering efficiency, effective customer price, task volume, service intensity and physical consumption. Tokens and calls can be supporting activity measures; they are insufficient proxies for comparable useful output on their own.

Identify an efficiency change and measure price pass-through. A phased rollout, randomized access or a credible comparison group can help distinguish its demand effect from general adoption. A before-and-after rise in resource use without a counterfactual is only descriptive.

For an energy claim, include the chosen IT or facility boundary consistently, alongside idle consumption, cooling overhead and utilization. Do not equate energy consumption with installed capacity or with emissions; emissions also depend on the energy supply.

Report uncertainty and the study's horizon. A short-run budget constraint can suppress demand that expands later. Conversely, an initial burst of experimentation can overstate persistent demand.

## 9. Confirmation, rejection and implications

The causal hypothesis gains support when an identified efficiency improvement induces additional demand that exceeds measured physical savings relative to a credible counterfactual. Decomposing that increase between N and I makes the result interpretable.

The backfire hypothesis is rejected for the studied setting if the induced response remains below the engineering saving with sufficiently precise measurement. That still allows partial rebound. A finding for one setting cannot reject or confirm a universal AI-wide law, which this paper does not propose.

A confirmed rebound may increase pressure on locally constrained infrastructure. It does not establish that all data-center owners benefit: financing, contractual allocation and operating costs determine who retains the value. The companion papers examine those separate questions.

## 10. Conclusion and limits

The Solland Paradox is a proposed AI-specific framing of established rebound economics. Its useful contribution is the separation of task count, service intensity and resource efficiency, together with a causal measurement requirement.

The paper derives a conditional test but supplies no original empirical estimate. Whether the condition holds, why it holds and how long it persists remain research questions. Efficiency and rising consumption can coexist; identifying efficiency as the cause requires more than observing both.

## References

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